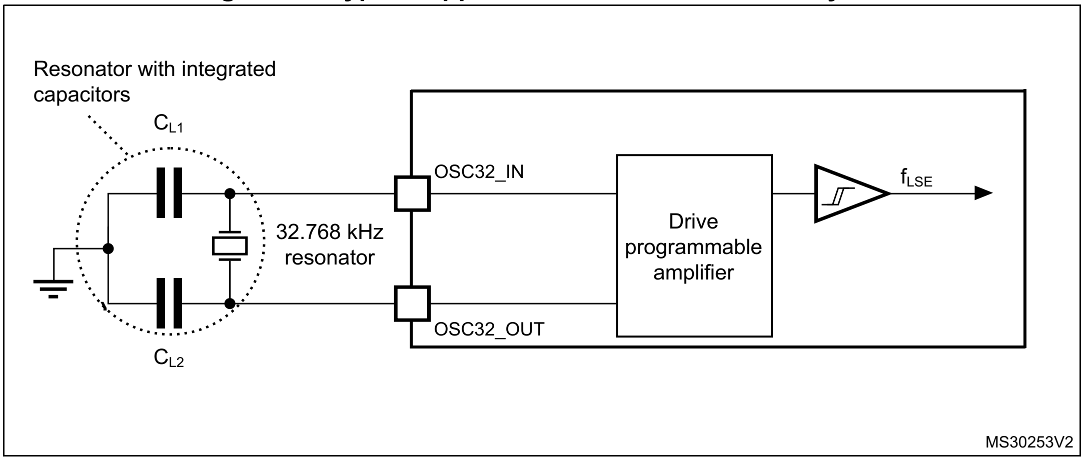
Note: For information on selecting the crystal, refer to the application note AN2867 "Oscillator design guide for ST microcontrollers" available from the ST website www.st.com.

Figure 20. Typical application with a 32.768 kHz crystal



Note: An external resistor is not required between OSC32_IN and OSC32_OUT and it is forbidden to add one.

5.3.8 Internal clock source characteristics

The parameters given in *Table 41* are derived from tests performed under ambient temperature and supply voltage conditions summarized in *Table 24: General operating conditions*. The provided curves are characterization results, not tested in production.

High-speed internal (HSI48) RC oscillator

Table 41. HSI48 oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSI48}	HSI48 Frequency	V_{DD}=3.0 V, T_A=30 °C	47.92	-	48.40	MHz
$\Delta_{Temp(HSI)}$(1)	HSI48 oscillator frequency drift over temperature and V_{DD} full voltage range	T_A= 0 to 85 °C	-1	-	1	%
		T_A= -40 to 125 °C	-2.5	-	2	%
TRIM(1)	HSI48 oscillator frequency user trimming step	From code 127 to 128	-8	-6	-4	%
		From code 63 to 64 From code 191 to 192	-5.8	-3.8	-1.8	
		For all other code increments	0.2	0.3	0.4	
D_{HSI48}(2)	Duty cycle	-	45	-	55	%
$t_{su(HSI48)}$(2)	HSI48 oscillator start-up time	-	-	1.4	1.8	µs
$t_{stab(HSI48)}$(2)	HSI48 oscillator stabilization time	at 1% of target frequency	-	1.5	3.6	µs
$I_{DD(HSI48)}$(1)	HSI48 oscillator power consumption	-	-	525	570	µA